Computational Physics Classes No.3 - Report

Simulation of radioactive decay: stiffness in ODE

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Abstract

The numerical simulation of a radioactive decay chain was performed using the implicit trapezoidal method with Newton iteration and adaptive time step control. The algorithm was tested for both non-stiff and stiff systems, showing excellent agreement with analytical solutions. The results demonstrate that adaptive step adjustment ensures stability and efficiency even for highly stiff differential equations.

1 Introduction

The purpose of this exercise is to study the numerical solution of a system of ordinary differential equations describing radioactive decay chains. Such systems often exhibit stiffness when decay constants differ by several orders of magnitude. To handle this difficulty, the implicit trapezoidal method combined with Newton iteration and an adaptive time step control was implemented. The obtained numerical results are compared with the exact analytical solutions to verify correctness and to demonstrate the effect of stiffness and tolerance on the stability and efficiency of the integration.

2 Description of the Numerical Method

The system of three coupled first-order differential equations describing radioactive decay was solved using the implicit trapezoidal method, which is a second-order accurate and unconditionally stable scheme. Because the method is implicit, the solution at each time step requires solving nonlinear equations, which was done using the Newton iteration method. To efficiently handle stiffness, the algorithm was combined with an adaptive time step control based on step-doubling and local error estimation. The step size was automatically adjusted to maintain the prescribed accuracy defined by the tolerance parameter.

3 Results

3.1 Implementation of the Adaptive Trapezoidal Scheme

The adaptive time step algorithm combined with the implicit trapezoidal method was successfully implemented in `C++`. The iterative Newton solver converged for each time step within the predefined tolerance of 10^{-10}, and the algorithm dynamically adjusted Δt according to the estimated local error. The implementation was verified to produce stable results even for stiff systems, demonstrating the robustness of the implicit approach.

3.2 Test of Correctness: $\lambda_0 = 1$, $\lambda_1 = 5$, $\lambda_2 = 50$, TOL $= 10^{-4}$

To verify the correctness of the numerical scheme, the system was solved for moderate stiffness parameters and compared with the analytical solutions. Figure 1 shows excellent agreement between the numerical and exact results for all components $N_0(t)$, $N_1(t)$, and $N_2(t)$. The decay chain evolves smoothly and the relative errors remain negligible over the entire simulation range.

The adaptive time step evolution is shown in Figure 2. Initially, small Δt values are used to capture the rapid decay of N_0, while later, as the system approaches equilibrium, the step size increases by several orders of magnitude. This confirms that the adaptive control efficiently balances accuracy and computational cost.

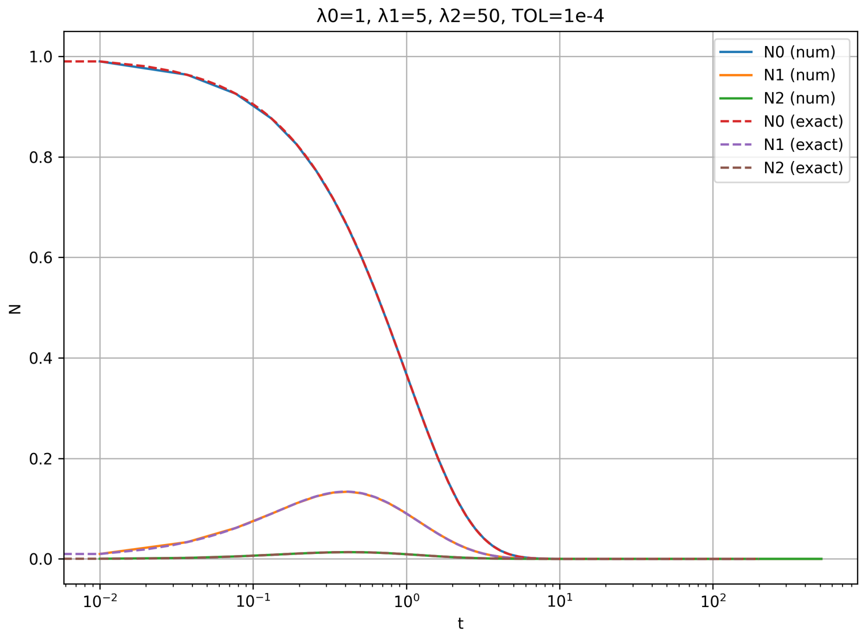

Figure 1: Comparison of numerical and exact solutions for $\lambda_0 = 1$, $\lambda_1 = 5$, $\lambda_2 = 50$, and TOL $= 10^{-4}$.

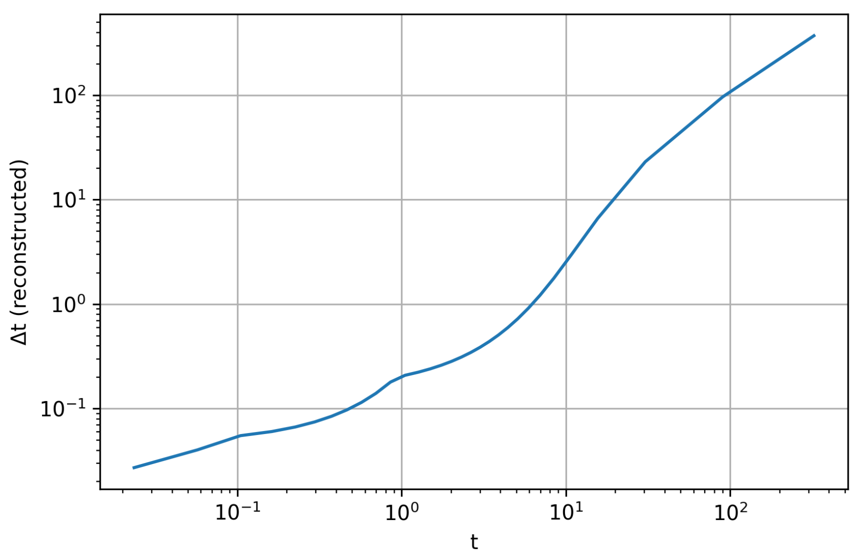

Figure 2: Adaptive time step evolution for $\lambda_0 = 1$, $\lambda_1 = 5$, $\lambda_2 = 50$, and TOL $= 10^{-4}$.

3.3 Stiff System: $\lambda_0 = 100$, $\lambda_1 = 1$, $\lambda_2 = 0.01$, TOL $= 10^{-6}$ and 10^{-3}

For the stiff configuration, where decay constants differ by several orders of magnitude, the algorithm was tested with two different tolerance levels. Figures 3–6 present the numerical results.

For TOL $= 10^{-6}$ (Figures 3 and 4), the numerical results closely match the analytical solutions. The adaptive step control keeps Δt very small at the beginning to resolve the fast decay of N_0, later gradually increasing the step as slower processes dominate. The time step varies over more than five orders of magnitude, demonstrating strong stiffness handling.

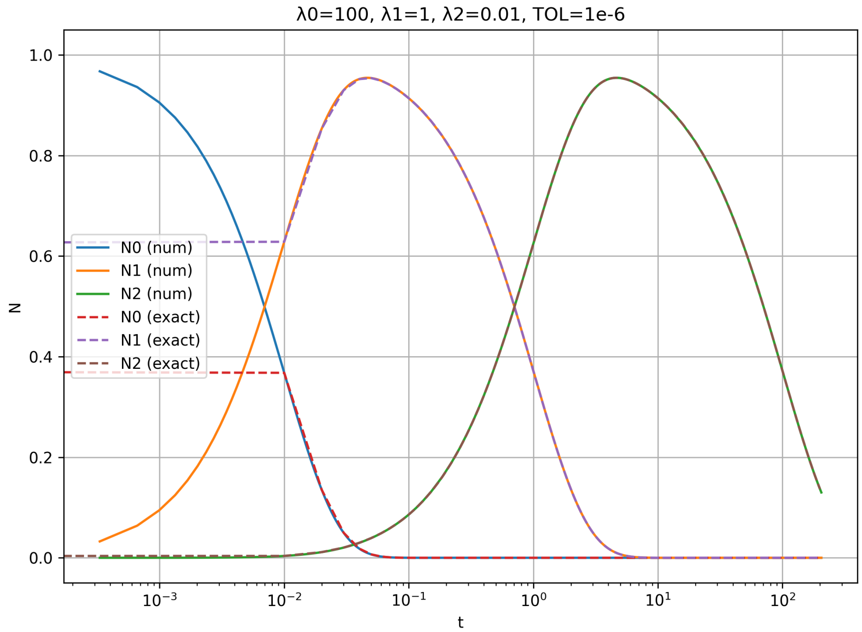

Figure 3: Numerical and exact solutions for the stiff system with $\lambda_0 = 100$, $\lambda_1 = 1$, $\lambda_2 = 0.01$, and $\mathrm{TOL} = 10^{-6}$.

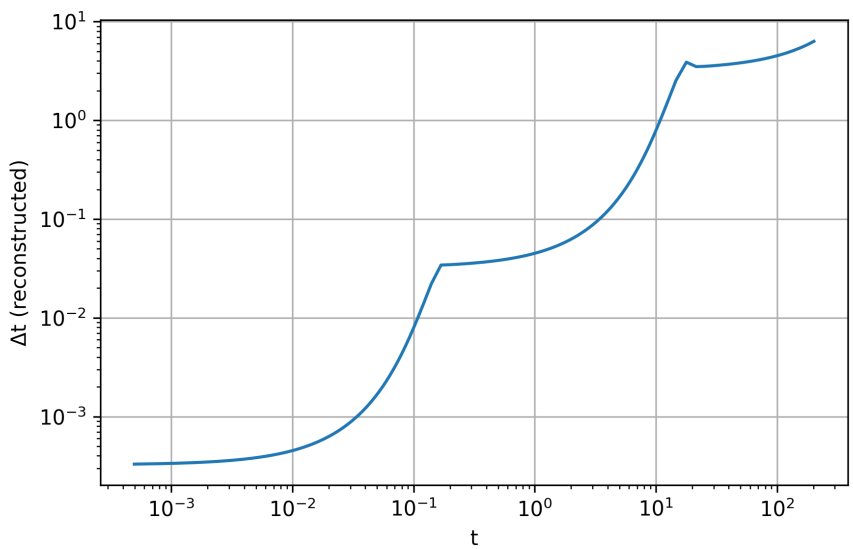

Figure 4: Adaptive time step evolution for $\mathrm{TOL} = 10^{-6}$.

For $\mathrm{TOL} = 10^{-3}$ (Figures 5 and 6), the time steps become significantly larger. As a result, while the general decay trends are preserved, minor deviations from the exact solution appear, especially for $N_2(t)$ at later times. The looser tolerance allows greater integration error, illustrating the trade-off between accuracy and computational efficiency.

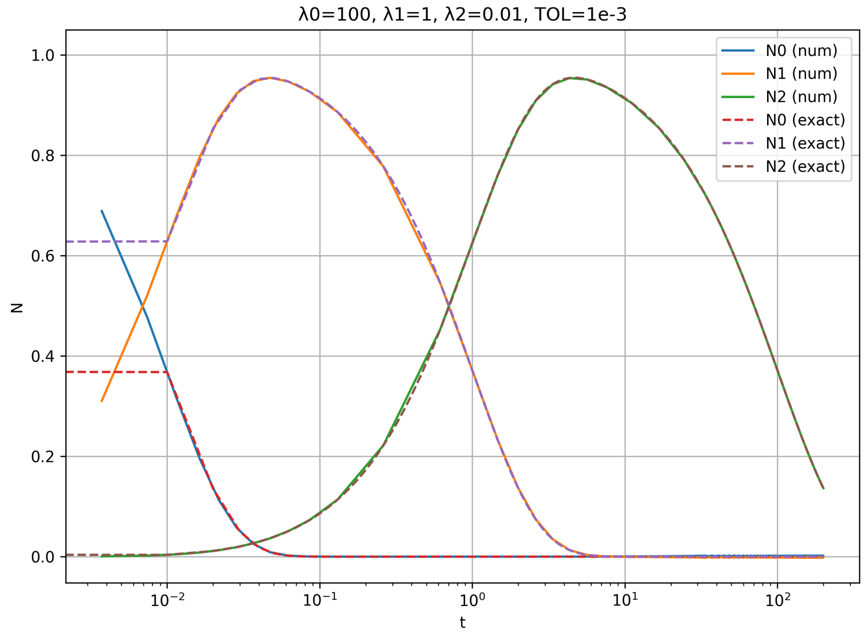

Figure 5: Numerical and exact solutions for $\mathrm{TOL} = 10^{-3}$.

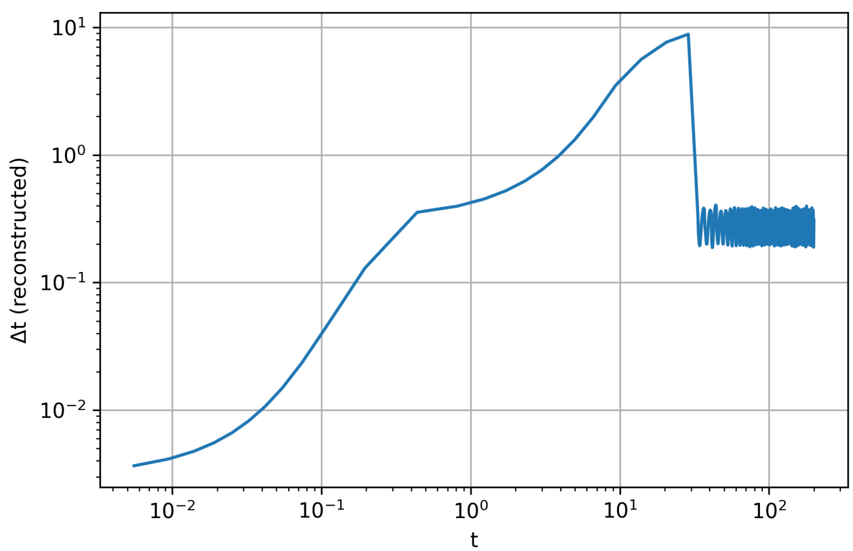

Figure 6: Adaptive time step evolution for $\mathrm{TOL} = 10^{-3}$.

The plot of the reconstructed adaptive time step (Figure 6) reveals that for loose tolerance ($\mathrm{TOL} = 10^{-3}$), the step size initially increases as the system evolves from fast to slow decay phases. However, at later times ($t \gtrsim 10^1$), Δt suddenly drops and begins to oscillate around a nearly constant value. This behavior is caused by the slow decay of $N_2(t)$, which remains non-negligible over a long timescale. The adaptive algorithm attempts to control the local error for this very small derivative, leading to repeated readjustments of Δt. Although the oscillations indicate sensitivity to rounding and error estimation at large t, the overall stability of the integration is maintained, confirming the robustness of the method even for stiff and slowly varying regimes.

3.4 Discussion of Tolerance Dependence

Comparing both tolerance cases, it is clear that a smaller tolerance (10^{-6}) results in finer time resolution and higher accuracy, while the larger tolerance (10^{-3}) yields a much coarser time grid. The discrepancy between the results is most visible in the slowest-decaying component $N_2(t)$, which is sensitive to accumulated numerical errors. This behavior highlights the importance of choosing tolerance values appropriate to the stiffness of the system and desired accuracy.

4 Conclusions

The implicit trapezoidal method combined with adaptive time stepping proved to be an effective and stable approach for solving stiff systems of ordinary differential equations describing radioactive decay chains. The numerical results showed excellent agreement with analytical solutions, confirming the correctness of the implementation. The adaptive step control efficiently adjusted the time step according to the local error, using small Δt during rapid transients and larger steps in slowly varying regions. For stiff cases, tighter tolerance values ensured higher accuracy at the cost of computational effort, while looser tolerances led to visible deviations, especially in the slowest-decaying component $N_2(t)$. Overall, the method successfully balanced accuracy, stability, and efficiency in the simulation of stiff decay systems.

References

[1] Dr. Hab. Eng. Chwiej, T. *Project: Simulation of radioactive decay: stiffness in ODE*. AGH University of Krakow. Retrieved from https://galaxy.agh.edu.pl/~chwiej/comp_phys/labs/4_radioactive_decay_stiff.pdf